

Program : Diploma in Computer Application and Business Management	
Course Code : 2252	Course Title: Business Mathematics
Semester : 2	Credits: 3
Course Category: Foundation Course	
Periods per week: 3 (L:3, T:0, P:0)	Periods per semester: 45

Course Objectives:

- To provide a thorough knowledge in Business Mathematics.
- To apply their knowledge to solve real world problems using the concepts of matrices, probability, commercial calculations and calculus.

Course Prerequisites:

Topic	Program/Course name
Basics of Mathematics	Secondary School

Course Outcomes:

On completion of the course, students will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Apply the concept of matrices and use them in real world problem solving.	10	Applying
CO2	Make use of probability concepts in various real life situations and apply them in decision making.	9	Applying
CO3	Use simple and compound interests, percentages, and profitability of business in business world problem solving and decision making.	13	Applying
CO4	Relate the basic concept of calculus with the field of business.	11	Applying
	Series Exam	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3	3	3				
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Apply the concept of matrices and use them in real world problem solving.		
M1.01	Recall the meaning of matrices, determinants of matrices & enumerate the properties of determinants, solve problems using these properties.	3	Understanding
M1.02	Classify types of matrices and apply various operations of matrices	4	Applying
M1.03	Recall transpose of matrix, illustrate and apply the concept of transpose of matrix.	3	Applying
Contents: Matrices - determinant of a square matrix-State the properties of determinants - Solve practical problems by applying these properties Differentiate different types of matrices - Define different operations of matrices - Equality - Addition of matrices - Subtraction - Multiplication by a scalar - Define the multiplication between two matrices. Do problems - State the condition for conformability for multiplication - Define the transpose of matrix - State theorems on transpose - Solve practical problems involving transpose of matrices.			
CO2	Make use of probability concepts in variousreal life situations and apply them in decision making.		
M2.01	Recall the meaning and understand the concept of Probability	2	Understanding
M2.02	Illustrate the meaning of random experiment, event , sample space	2	Understanding
M2.03	Classify different types of events.	2	Understanding
M2.04	Apply the concepts of Probabilities by problem solving.	3	Applying
	Series Test – I	1	

Contents: Probability - Define random experiment - Define event, sample space - Define different types of events - Explain mutually exclusive events - Explain equally likely events - Explain Exhaustive events - Explain Independent events - Explain Dependent events.			
CO3	Use simple and compound interests, percentages, and profitability of business in business world problem solving and decision making.		
M3.01	Illustrate the concept Simple Interest. Calculate Principal, Rate, Time & Interest amount as per given data	3	Applying
M3.02	Illustrate the concept Compound Interest Calculate Principal, Rate, Time & Interest amount as per given data	3	Applying
M3.03	Illustrate the concept of Time value of Money.	1	Understanding
M3.04	Understand the concept of Percentage and its application for calculating of percentage of various real world situations.	3	Applying
M3.05	Understand and Apply the concept of Profit and Loss and its percentage for calculating profitability of various commercial activities	3	Applying
Contents: Simple Interest - Calculate the Interest Amount - Principal - Rate - Time period as per the data provided. Compound Interest - Calculate the Compound Interest Amount - Principal - Rate-Time period as per the data provided. Time value of Money - Illustrate the concept of Time value of Money. Percentage - Meaning of Percentage - Calculation of percentage, regarding, marks, population, money, income etc. Profit and Loss - Calculation of profit or loss and percentage profit or loss - Calculations of cost price and selling price - Total profit or loss while buying & selling different articles at different rates.			
CO4	Relatethe basic concept of calculus with the field of business.		
M4.01	Understand the concept differentiation and differentiation from first principle-Solve problems using first principle	2	Understanding
M4.02	Calculate the derivative of standard forms.	2	Applying
M4.03	Derive basic theorems of differentiation and Deduce product rule, quotient rule of differentiation.	2	Applying
M4.04	Understand the Concept of Integration and enumerate fundamental results of Integration.	2	Understanding
M4.05	Calculate the integrals of standard forms and integration of products and quotients	2	Applying

M4.06	Application of Calculus in Business and Economics.	1	Understanding
	Series Test – II	1	
Contents: Differentiation - State differentiation from first principle - Solve problems using first principle - Find the derivative of standard forms - Derive basic theorems of differentiation - Deduce product rule, quotient rule of differentiation - Solve problems using the above rule. Integration - State fundamental results of Integration - Find the integrals of standard forms - Find Integration of products and quotients - Application of Calculus in Business and economics			

Text / Reference:

T/R	Book Title/Author
R1	J.K Singh, Business Mathematics , Himalaya publication Pvt. House Ltd.,2019
R2	B.M. Agarwal, Business Mathematics & Statistics , Ane's Student Edition,1E
R3	S Rajagopalan and R Sathanathan, Business Mathematics , VNI 3rd edition
R4	Dr. Padmalochan Hazarika, A Text book of Business Mathematics , S Chand Publishing, 4 th edition

Online Resources:

Sl.No	Website Link
1	https://www.mathsisfun.com/algebra/matrix-introduction.html
2	https://www.tutorialspoint.com/statistics/probability.htm
3	https://youtu.be/Je20KPOFPnI
4	https://wordpandit.com/simple-interest-basic-concept/